

Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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Topic: Circle Theorems

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

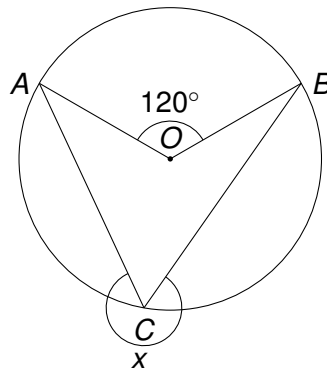
Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** A radio telescope dish is mounted on a circular track of centre O , used to track satellites passing overhead. Points A and B lie on the circle, marking two sensor positions, and C is a third sensor position on the circumference.

O is the centre of the circle. The angle at the centre, $\angle AOB = 120^\circ$.



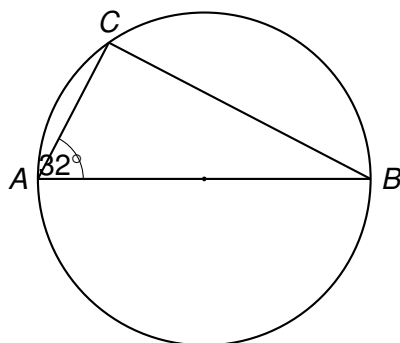
Find the size of angle x , the angle at the circumference at point C .

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 1 is 2 marks)

- 2 A team of space engineers is designing a circular viewing window for a satellite. AB is a diameter of the circular window, and C is a point on the circumference where a support strut meets the glass, forming triangle ABC .



AB is a diameter of the circle. Angle $BAC = 32^\circ$.

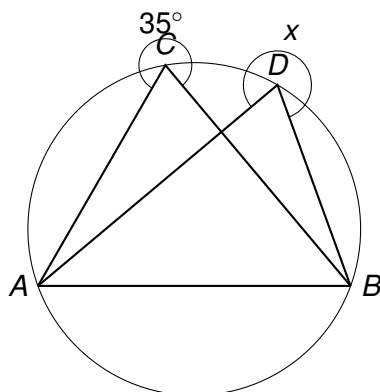
Work out the size of angle ACB , marked x on the diagram, giving a reason for your answer.

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 2 is 2 marks)

- 3 An astronomer plots the positions of a satellite's dish reflector and two ground observation stations on a circular diagram representing the horizon, with points A , B , C and D marked on the circle. Chord AB is shared by triangles ACB and ADB , where C and D lie on the same arc on the same side of AB .



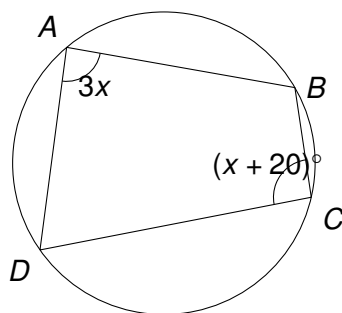
Points A , B , C and D lie on the circle. Angle $ACB = 35^\circ$.
Find the size of angle x , giving a reason for your answer.

$$x = \dots\dots\dots^\circ$$

(3)

(Total for Question 3 is 3 marks)

- 4 An archaeologist is studying the circular floor plan of an ancient observatory once used to track the movement of stars. Four marker stones A , B , C and D lie on the circular boundary wall, forming a cyclic quadrilateral $ABCD$.



Angle $DAB = 3x^\circ$ and angle $BCD = (x + 20)^\circ$.

- (a) Form an equation in x and solve it to find the value of x .

$$x = \dots\dots\dots$$

(2)

- (b) Work out the size of angle DAB .

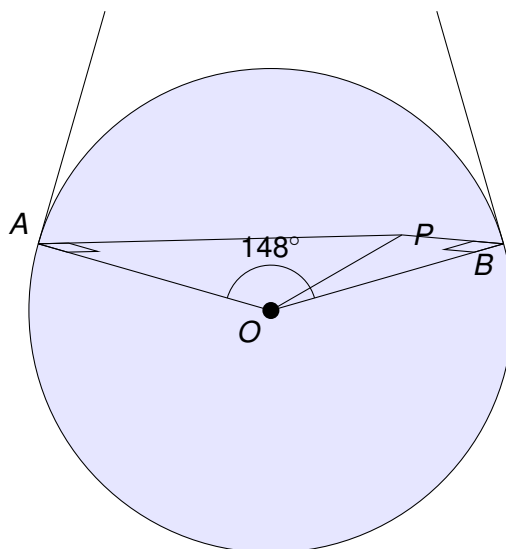
$$DAB = \dots\dots\dots^\circ$$

(2)

(Total for Question 4 is 4 marks)

- 5 A satellite dish tracking station is represented by the external point P , from which two signal paths PA and PB are drawn tangent to a circular exclusion zone with centre O , touching the circle at points A and B .

The angle $AOB = 148^\circ$, as shown in the diagram.



Calculate the size of angle APB .

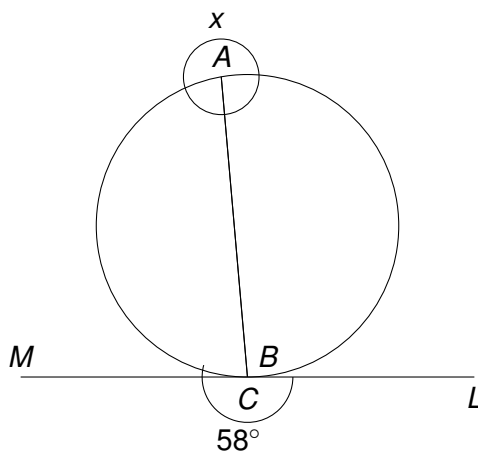
$$\angle APB = \dots\dots\dots^\circ$$

(3)

(Total for Question 5 is 3 marks)

- 6 A satellite dish support ring has a circular cross-section. A signal receiver arm is bolted to the ring at point C , running along the tangent line LM to the circle at that point. Two internal struts, CA and CB , are fixed as chords from C to two other mounting points, A and B , on the ring, with strut AB completing a triangular brace inside the dish.

The angle between the tangent (receiver arm) LM and the chord (strut) CB is 58° , as shown.



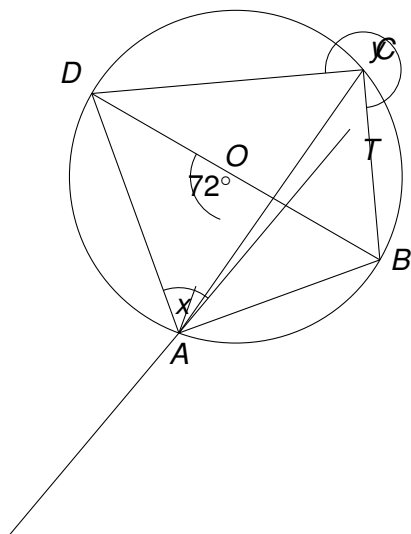
Using the alternate segment theorem, find the size of angle x , the internal support angle at A between struts AC and AB .

$$x = \dots\dots\dots^\circ$$

(4)

(Total for Question 6 is 4 marks)

- 7 A historian studying ancient astronomical instruments encounters a circular sundial engraved with a diagram used to mark key angles. Points A , B , C and D lie on the circumference of a circle with centre O . TA is a tangent to the circle at A . BOD is a straight line through the centre.



Given that $\angle DOA = 72^\circ$, show that $\angle BCD = 106^\circ$.

You must show each step of your working and give a geometric reason at every stage.

(6)

(Total for Question 7 is 6 marks)

TOTAL FOR PAPER IS ?? MARKS